

Imperfect Competition I: Monopoly

1 Monopoly

1. The monopolist's problem

$$\max_q \Pi \equiv P(q)q - C(q) = R(q) - C(q).$$

2. The profit-maximization condition for monopolist

$$MR(q) = MC(q).$$

3. The condition that $MR = MC$ is also true in perfect competition. The difference is :

(a) In perfect competitive market: $P = MR = MC = AC$, that is a firm's marginal revenue equals the market price, marginal cost equals average cost and profit is zero.

(b) In monopoly market, it is usually the case that $MR < P$.

4. Inverse elasticity pricing rule

(a) Elasticity

$$\epsilon(q) = \frac{dq}{dP} \frac{P}{q}$$

(b) Inelastic:

$$|\epsilon(q)| \in (0, 1),$$

(c) Unitary elastic

$$|\epsilon(q)| = 1,$$

(d) Elastic

$$|\epsilon(q)| > 1.$$

(e) Marginal revenue and price elasticity

$$\begin{aligned} MR(q) &= P + q \frac{dP}{dq} \\ &= P \left(1 - \frac{1}{|\epsilon(q)|} \right). \end{aligned}$$

$MR(q)$ is positive if and only if $|\epsilon(q)| > 1$.

(f) Inverse elasticity rule

Profit-maximizing condition

$$MR = MC \implies \frac{P - MC}{P} = \frac{1}{|\epsilon(q)|}.$$

5. Market power

(a) Market power is the capacity of a firm to control over its price.

(b) Lerner Index of market power

$$\text{Index} = \frac{P - MC(q)}{P(q)} = \frac{1}{|\epsilon(q)|}.$$

(c) In perfect competition: $P = MR = MC$, Index = 0. Monopoly:
 $P > MR = MC$, Index > 0.

2 Price Discrimination

2.1 First Degree

1. In this case, the monopoly perfectly observes the demand functions of all the consumers and therefore can appropriate consumer surplus as a whole.
2. One practice is the two-part tariff such that the unitary price is the competitive price and franchise A_i applied to consumer i is consumer surplus.

2.1.1 Two-Part Tariffs

1. The two-part tariff is thus

$$\begin{cases} A + pq & \text{if } q > 0 \\ 0 & \text{if } q = 0 \end{cases}$$

where A is the access fee, and p is the unit price.

2. The monopoly can therefore choose to tariff $p = p^c$ and $A = CS = \int_{p^c}^k D(p)dp$.
3. All social surplus is then appropriated to the monopoly, which realizes the maximum of its profit, but the social distortion disappears.

2.2 Second Degree

1. Assume that there are two types of consumers, $i = 1, 2$, and that consumer i has as his utility $\theta_i q_i - p_i$, where θ_i is a taste parameter. Suppose that $\theta_2 > \theta_1$, so that consumer 2 is more eager for the good than consumer 1.
2. If the monopoly could observe each consumer's type, it would solve for each type

$$\max_{\{p_i, q_i\}} [p_i - C(q_i)] \quad \text{s.t.} \quad \theta_i q_i - p_i \geq 0$$

This gives the first best solution (q_i^*, p_i^*) such that $\theta_i = C'(q_i^*)$, $p_i^* = \theta_i q_i^*$. This is the first degree case.

3. In the second degree case, each consumer knows of course his own type but the monopoly does not know. What the monopoly knows is the proportion of the type 2 in the population is μ .
4. The monopoly therefore offers a direct revealing mechanism: two contracts (p_1, q_1) and (p_2, q_2) such that the type 1 consumer chooses the first and type 2 the second.
5. The program the monopoly solves is then

$$\max_{\{p_1, q_1; p_2, q_2\}} \{\mu[p_2 - C(q_2)] + (1 - \mu)[p_1 - C(q_1)]\}$$

subject to

$$\theta_1 q_1 - p_1 \geq \theta_1 q_2 - p_2 \quad (\text{IC}_1)$$

$$\theta_2 q_2 - p_2 \geq \theta_2 q_1 - p_1 \quad (\text{IC}_2)$$

$$\theta_1 q_1 - p_1 \geq 0 \quad (\text{IR}_1)$$

$$\theta_2 q_2 - p_2 \geq 0 \quad (\text{IR}_2)$$

6. ICs are the incentive constraints that express that each type must voluntarily choose the contract meant for him. IRs are participation constraints which guarantee to each type a nonnegative utility.

7. IR_1 is active or binding and IR_2 can be neglected. Proof: By IC_2 , we obtain

$$\theta_2 q_2 - p_2 \geq \theta_2 q_1 - p_1 \geq \theta_1 q_1 - p_1$$

If IR_1 is not binding, so would be IR_2 , and we could increase p_1 and p_2 . This would increase the Principal's profit without any effect on IC.

8. IC_2 is active. Proof: If IC_2 is not active, then

$$\theta_2 q_2 - p_2 > \theta_2 q_1 - p_1 \geq \theta_1 q_1 - p_1 = 0$$

We can therefore increase p_2 without breaking IC or IR_2 .

9. $q_2 \geq q_1$. Proof: Adding IC_1 and IC_2 gives

$$\theta_2(q_2 - q_1) \geq \theta_1(q_2 - q_1).$$

10. IC_1 can be neglected. Proof: Since IC_2 is active and $q_2 \geq q_1$,

$$p_2 - p_1 = \theta_2(q_2 - q_1) \geq \theta_1(q_2 - q_1).$$

11. Therefore, the above program remains as

$$\max_{\{p_1, q_1; p_2, q_2\}} \{\mu[p_2 - C(q_2)] + (1 - \mu)[p_1 - C(q_1)]\}$$

subject to

$$\begin{aligned} \theta_2 q_2 - p_2 &= \theta_2 q_1 - p_1 \\ \theta_1 q_1 - p_1 &= 0 \end{aligned}$$

12. After replacing p_1 and p_2 by their values, we obtain

$$\max_{\{q_1, q_2\}} \{\mu[\theta_1 q_1 + \theta_2(q_2 - q_1) - C(q_2)] + (1 - \mu)[\theta_1 q_1 - C(q_1)]\}$$

Now we easily get

$$\begin{aligned} C'(q_2) &= \theta_2 \\ C'(q_1) &= \theta_1 - \frac{\mu}{1 - \mu}(\theta_2 - \theta_1) \end{aligned}$$

13. The results shows that $q_2^{**} = q_2^*$, $q_1^{**} < q_1^*$. And $p_1 = \theta_1 q_1^{**}$, $p_2 = \theta_2 q_2^* - (\theta_2 - \theta_1) q_1^{**}$, where $(\theta_2 - \theta_1) q_1^{**}$ is so-call "the information rent."
14. The monopoly must prevent type 2 from masquerading as type 1 in order to pay the smaller price cosigned to type 1's. To succeed in this, the monopoly will sell a smaller quantity to type 1, which makes the lower price less interesting to type 2.
15. This is the famous "screening" or "adverse selection" problems. The self-selection mechanisms or menus have been applied to various problems such as optimal regulation, auctions, taxation, insurances, etc.

2.3 Third Degree

1. Assume that there are m groups of consumers, $i = 1, \dots, m$, distinguishable by observable characteristics: the demand of group i is a function $D_i(p)$ known to the monopoly.
2. It can therefore tariff distinct prices $p_1, \dots, p_m$ that maximize the total profit

$$\sum_{i=1}^m p_i D_i(p_i) - C\left(\sum_{i=1}^m D_i(p_i)\right)$$

We immediately get the classical formula, but this time good by good:

$$\forall i = 1, \dots, m, \quad \frac{p_i - C'}{p_i} = \frac{1}{|\epsilon_i(p)|}$$

where ϵ_i is the demand elasticity of group i . It is necessary to make groups with the least elastic demand pay more.

3 Durable goods

1. Take the monopoly sells a durable goods. It can be thought compete with itself, since consumers may buy the good at the moment when they think it the least expensive.
2. Consider a model with two periods, $t = 1, 2$, where the demand for services provided by the durable good is given by $D(p) = 1 - p$ in each period. Suppose that the good is produced at no cost, and the discount factor is δ .
3. If the monopolist rents the durable good, at each period it will maximize the profit $p(1 - p)$, whence $p_1 = p_2 = \frac{1}{2}$ and a discounted profit $\frac{1+\delta}{4}$.

4. When the monopoly decides to sell the durable good, the consumers must choose when to purchase. It is necessary to apply backward induction to reason starting from the second period.
5. Suppose that the quantity q_1 had been sold in the first period. The residual demand at the beginning of the second period is $1 - q_1 - p$. The monopoly must then choose the quantity it sells, q_2 , so as to maximize $q_2(1 - q_1 - q_2)$, whence $p_2 = q_2 = \frac{1-q_1}{2}$ and a second period profit $\pi_2 = \frac{(1-q_1)^2}{4}$.
6. Back to the first period. Since the buyer knows that he can resell the good in the second period at the price p_2 , he is ready to pay δp_2 more than the normal price $1 - q_1$. We get

$$p_1 = 1 - q_1 + \delta p_2 = (1 - q_1)(1 + \frac{\delta}{2})$$

The discounted profit is therefore

$$q_1(1 - q_1)(1 + \frac{\delta}{2}) + \delta \frac{(1 - q_1)^2}{4}$$

7. We get $q_1 = \frac{2}{4+\delta}$ and a discounted profit

$$\pi = \frac{(2 + \delta)^2}{4(4 + \delta)} < \frac{1 + \delta}{4}$$

which shows that the monopoly's profit is smaller when the good is sold than when it is rented.

8. In effect a monopoly that sells the durable good cannot commit itself not to lower the price of the good later; this encourages consumers to wait. As a matter of fact, we can show in our model that $p_1 > p_2$ so that the monopoly effectively lowers its prices in the course of time.
9. One intuitively senses that the monopoly's profit is smaller because it has more difficulty committing to a price policy. For that matter, Coase has conjectured that if the periods were cut into shorter and shorter subperiods and if corresponding δ tends to 1, then the monopoly's profit will tend towards 0. In the limit the monopoly's power disappears completely. This is so-called Coase Conjecture and was proved at the beginning of the 1980s.

4 Natural Monopoly

1. The cost function is subadditive if

$$\forall n, \forall i = 1, \dots, n, C\left(\sum_{i=1}^n q_i\right) \leq \sum_{i=1}^n C(q_i)$$

and $n = 1$ at the optimum. It would be better if a single firm produced, rather than production being shared among several firms. It is then said that the sector in question is a natural monopoly.

2. Habitually cited natural monopoly are transportation, telecommunication, water, electricity and so on. All these sector operate with high fixed costs or heavy infrastructure. For example, it seems that even if competition is desirable, it would be absurdly costly to double the number of railroad lines.
3. For purely technology reasons, it is fitting to grant a monopoly to one of the firms. However, the firm may then behave as a monopoly, which is socially harmful as we indicate above.
4. The problem is therefore how this type of sector must be regulated. The marginal cost pricing suggest that let $p = MC$, this is the efficiency condition. However, the natural monopoly will make a loss since its average cost is always higher than the marginal cost. E.g. A firm with cost function $C(x) = F + cx$ will lose its fixed costs.
5. Some suggests replace marginal cost pricing by the average cost pricing, then the firm makes zero profit by construction. However, it is not optimal. Then the choice of pricing then becomes a second-best problem.

4.1 Asymmetric Information Regulation

1. Consider a natural monopoly with cost function $C(q, \theta) = K + \theta q$. The firm is given a subsidy t levied on the consumer, the social unit cost of which is ε : when the government collect a tax t , the utility of the consumer decreases by $(1 + \varepsilon)t$ because of tax management costs and the economic distortions which accompany taxes.
2. The firm's profit is therefore

$$t - C(q, \theta)$$

while the consumer surplus is

$$S(q) - (1 + \varepsilon)t = \int_0^q P(x)dx - (1 + \varepsilon)t$$

where P is the inverse demand function.

3. We assume the regulator's objective is social welfare, or the non-weighted sum of the consumer surplus and of firm profit

$$S(q) - C(q, \theta) - \varepsilon t.$$

4.1.1 First-best

1. Suppose that the regulator observes θ . Then it would suffice him to solve

$$\max_{\{t, q\}} [S(q) - C(q, \theta) - \varepsilon t] \quad \text{s.t.} \quad t - C(q, \theta) \geq 0$$

whose solution is of course $t = C(q, \theta)$ and $S'(q) = P(q) = (1 + \varepsilon)\theta$, that is we see again the marginal cost pricing rule, corrected here for the cost of public funds ε .

4.1.2 Second-best

1. In fact it is more realistic to suppose that information is asymmetric: the firm obviously knows θ , but the regulator simply has a Bayesian prior such that θ follows a cumulative distribution function F with density f on an interval $[\underline{\theta}, \bar{\theta}]$.
2. The regulator must find a direct mechanism $(q(\theta), t(\theta))$ such that

$$\max \int_{\underline{\theta}}^{\bar{\theta}} \{S[q(\theta)] - C[q(\theta), \theta] - \varepsilon t(\theta)\} dF(\theta)$$

subject to

$$\forall \theta, \theta', \quad t(\theta) - C[q(\theta), \theta] \geq t(\theta') - C[q(\theta'), \theta] \quad (\text{IC})$$

$$\forall \theta, \quad t(\theta) - C[q(\theta), \theta] \geq 0 \quad (\text{IR})$$

3. Denote by $r(\theta) = t(\theta) - C[q(\theta), \theta] = t(\theta) - \theta q(\theta) - K$ the information rent of the firm θ . This is the profit that it succeeds in obtaining thanks to its private information. The IC can be rewritten

$$\forall \theta, \theta', \quad r(\theta) \geq r(\theta') - (\theta - \theta')q(\theta') \quad (\text{IC})$$

By exchanging the role of θ and θ' , we get a second inequality that can be combined with the first. It is written as

$$\forall \theta, \theta', (\theta - \theta')q(\theta) \leq r(\theta') - r(\theta) \leq (\theta - \theta')q(\theta')$$

which implies that $(\theta - \theta')(q(\theta') - q(\theta)) \geq 0$ and therefore q is decreasing.

4. By assume that $\theta \leq \theta'$, and by divided by $(\theta - \theta')$, we get

$$-q(\theta) \leq \frac{r(\theta') - r(\theta)}{\theta' - \theta} \leq -q(\theta')$$

So at every point where q is continuous, r may be differentiated and $r' = -q$. We can then deduce that r is convex since q is decreasing.

5. Since q is positive, r decreases, and the constraint of participation amount to just $r(\bar{\theta}) = 0$. Therefore we get

$$r(\theta) = \int_{\theta}^{\bar{\theta}} q(t)dt$$

which gives the transfers t in terms of q :

$$t(\theta) = C[q(\theta), \theta] + \int_{\theta}^{\bar{\theta}} q(t)dt$$

6. Having eliminated the transfers, it only remains to

$$\max_{\underline{\theta}} \int_{\underline{\theta}}^{\bar{\theta}} \{S[q(\theta)] - (1 + \varepsilon)\theta q(\theta) - \varepsilon \int_{\theta}^{\bar{\theta}} q(t)dt\} dF(\theta)$$

subject to

$$q'(\theta) \leq 0$$

7. We first neglect the constraint. The optimum is then found by integrating the objective by parts, whence

$$\int_{\underline{\theta}}^{\bar{\theta}} \{S[q(\theta)] - (1 + \varepsilon)\theta q(\theta) - \varepsilon \frac{F(\theta)}{f(\theta)} q(\theta)\} dF(\theta)$$

This result is simply maximized point by point, whence finally

$$S'[q(\theta)] = P[q(\theta)] = (1 + \varepsilon)\theta + \varepsilon \frac{F(\theta)}{f(\theta)}$$

8. If $F(\theta)/f(\theta)$ is an increasing function, which is for many usual distribution, then the right-hand side is increasing, and since P is decreasing, the solution $q(\theta)$ verifies the constraint $q'(\theta) \leq 0$.
9. Therefore the price this time is higher than the social marginal cost; the supplementary term is used to compel the firm to reveal its type. It leads to underproduction: q is smaller than the first-best. The regulator must keep the θ -type firm from posing as a less efficient firm $\theta' > \theta$ and thus from pocketing a higher subsidy. The most economic way to do this is to reduce the production of inefficient firms, so that they get smaller subsidies, and more efficient firms are not tempted to lie.

4.2 Deregulation

1. The regulation of natural monopolies has greatly evolved over the last few years as many firms were privatized and often faced fierce competition. Deregulation is based on the following beliefs:
 - (a) Technologies innovations make natural monopolies more expendable: even where the fixed costs remain high at the infrastructure level, infrastructures can be separated from the actual service activity, where returns to scale are rarely increasing.
 - (b) The diversified product range of these sectors (one only need to think of the new telecommunication products), makes detailed regulation problematic.
 - (c) Competition in itself is an effective regulator.
2. Demsetz (1968) argued that rather than regulate the distribution of utilities like electricity, the right to distribute electricity could be put up for auction and then granted to the highest bidder. In this way the monopoly rents would transfer to the taxpayer who could select the most efficient operator.
3. Price-cap type of regulation formula is adopted.